Data Analysis Project
Final Handin
(Group 05)

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1 Introduction

2 Integration

3 API

For the API, FastAPI was used to implement a REST interface for accessing the precipitation data stored in the german_prec table. The API provides basic reading (GET), adding (POST), modifying (PUT) and deleting (DELETE) functionality. The service connects directly to the database using psycopg2.
The API can be started locally using:

```
python -m uvicorn main:app --reload
```

Alternatively, it can be started via the provided `run_fastapi.bat` file.

After startup, the interactive API documentation is available at:

`http://127.0.0.1:8000/docs`

3.1 Data Model

All requests that insert or update data use the same JSON structure, defined using a Pydantic model. Each record is uniquely identified by a composite primary key consisting of the station_id and the station_timestamp. All remaining attributes are optional.

Example JSON object:

```
{
  "station_id": 164,
  "station_timestamp": "2024-09-01T00:00:00",
  "qn_8": 3,
  "precipitation_mm": 0,
  "precipitation_t_or_f": false,
  "precipitation_type": 0,
  "eor": "eor"
}
```

3.2 GET: Read Precipitation Data

Endpoint:

`GET /precipitation/{station_id}`

This endpoint returns up to 100 precipitation records for a given weather station. The limit is applied to prevent excessive data transfer and ensure acceptable response times.

Example request:

`GET /precipitation/164`

Example response:


```
[
  {
    "station_id": 164,
    "station_timestamp": "2024-09-01T00:00:00",
    "qn_8": 3,
    "precipitation_mm": 0,
    "precipitation_t_or_f": false,
    "precipitation_type": 0,
    "eor": "eor"
  }
]
```


3.3 POST: Insert New Data

Endpoint:

`POST /precipitation`

This endpoint inserts a new precipitation record into the database.

Example request body:


```
{
  "station_id": 1,
  "station_timestamp": "2024-09-01T00:00:00",
  "qn_8": 3,
  "precipitation_mm": 0,
  "precipitation_t_or_f": false,
  "precipitation_type": 0,
  "eor": "eor"
}
```


Example response:


```
{
  "message": "new tuple created"
}
```

3.4 PUT: Update Existing Data

Endpoint:

`PUT /precipitation/{station_id}/{station_timestamp}`

This endpoint updates an existing precipitation record identified by station ID and timestamp. The primary key itself cannot be modified.

Example request:

`PUT /precipitation/1/2024-09-01T00:00:00`

Example request body:

```
{
  "station_id": 1,
  "station_timestamp": "2024-09-01T00:00:00",
  "qn_8": 3,
  "precipitation_mm": 5,
  "precipitation_t_or_f": true,
  "precipitation_type": 1,
  "eor": "eor"
}
```

Example response:

```
{
  "message": "updated"
}
```

3.5 DELETE: Remove Data

Endpoint:

`DELETE /precipitation/{station_id}/{station_timestamp}`

This endpoint deletes a precipitation record using station ID and timestamp as identifiers. In the following example, the artificially created entry from the POST example is removed.

Example request:

`DELETE /precipitation/1/2024-09-01T00:00:00`

Example response:

```
{
  "message": "deleted"
}
```

4 Analysis

HOW TO (Remove this whole section !!!)

The purpose of this document structure is to ensure compatibility between all documents across all groups. Therefore, you are limited to this template. Our goal is to create a single document across all *p3 Project Reports*.

This template is used for the student reports of the Databases course. Remove this section by editing the `reportContent.tex` file **after** you have understood how to use this template.

Note: If you use this file as your template, make sure to replace `\section*{}` and `\subsection*{}` with `\section{}` and `\subsection{}` respectively. We did purposely use the non-numbered sections in this guide, however in your report the numbered sections are better suited.

Folder Structure

```
proceedings/
  |- reportContent/  <-- YOUR working folder
      |- images/     <-- Images are stored here
      |- sections/   <-- Place sections/tex-files here
          |- introduction.tex
          |- report.bib
          |- lessons-learned.tex
          '- section2.tex
      '- reportContent.tex  <-- Change section imports here
  |- proceedings.tex  <-- DO NOT alter this file
  |- _HOWTO_.tex     <-- *this* How To section
  |- llncs.cls                              <-- NOR that
  |- proceedings.py                         <-- NOR that
  |- splncs03.bst                           <-- NOR that
  '- UniBas_Logo_EN_Schwarz_RGB_65.pdf      <-- NOR that
```

Setup

Your report is compiled using the `proceedings.tex` file. Therefore, we suggest that you set the file `proceedings.tex` as the Master-/Root-Document that you can conveniently work on your report. Please be aware that you have to compile the `proceedings.tex` file in the following order:

1. `pdflatex` proceedings.tex
2. `biber` proceedings
3. `pdflatex` proceedings.tex
4. `pdflatex` proceedings.tex

For your convenience, you can compile this document by using the `proceedings.py` utility:

```
$> python proceedings.py -p
```

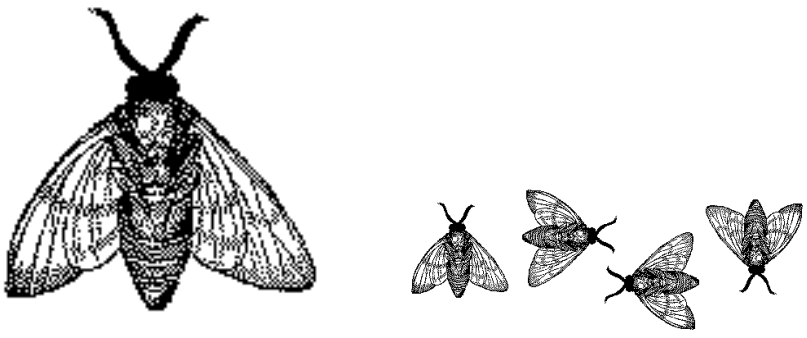
And subsequently (especially before your submission) clean the auxiliary LaTeX files:

```
$> python proceedings.py -z
```

Additionally, this template is compatible with Overleaf. You can simply upload the zip as a new project.

Images

Images placed in the `images` sub-folder can be used directly in the `\includegraphics` command – see also Figure 1. Also, sub-figures are possible: Figures 2a and 2b. Please DO NOT store your images elsewhere!
Please use your group number (e.g., *1*) in every image file name (e.g., *g1-rosette.pdf* instead of *rosette.pdf*)!
It is mandatory, that you do *only* specify the filename, as the path is set automatically. Hence, for Figure 1, the command to include the image is `\includegraphics{g1-rosette}`



(a) A fly (b) Flies

Fig. 2: Example figure stored in the `images` sub-folder

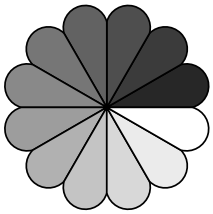


Fig. 1: Example figure stored in the `images` sub-folder and displayed with `\includegraphics[width=0.45\textwidth]{g1-rosette}`

Code

To illustrate code use the `lstlisting` environment:

```
public void foo( final int bar ) {
  System.out.println( bar );
}
```

Listing 1.1: Example for Java

```
x = 1
if x == 1:
    print("x is 1.")
```

Listing 1.2: Example for Python

```
SELECT * FROM users;
```

Listing 1.3: Example for SQL

External PDF

One can also easily import a PDF document using

`\includepdf[pages=1,pagecommand={\pagestyle{headings}}]{filename}`

to include the `filename.pdf` document. It is important to use the pagestyle *fancy*, so that the header is also on the included page. Furthermore, please ensure that your PDF version (or image) of the ER diagram is **readable**.

Labels and References

The proceedings template includes the package *cleveref*, which enables pretty references. One such reference is the following one: Figure 2, which was produced using the command `\Cref{fig:example subfigure}`. To use the full potential of such clever referencing, one have to set up labels, such as in the example before. The label command was `\label{fig:example_subfigure}` (as seen in the source code in line 69). Be aware that you have to prefix every single label you use with your group name, i.e. `g1` to not clash with other group's labels.

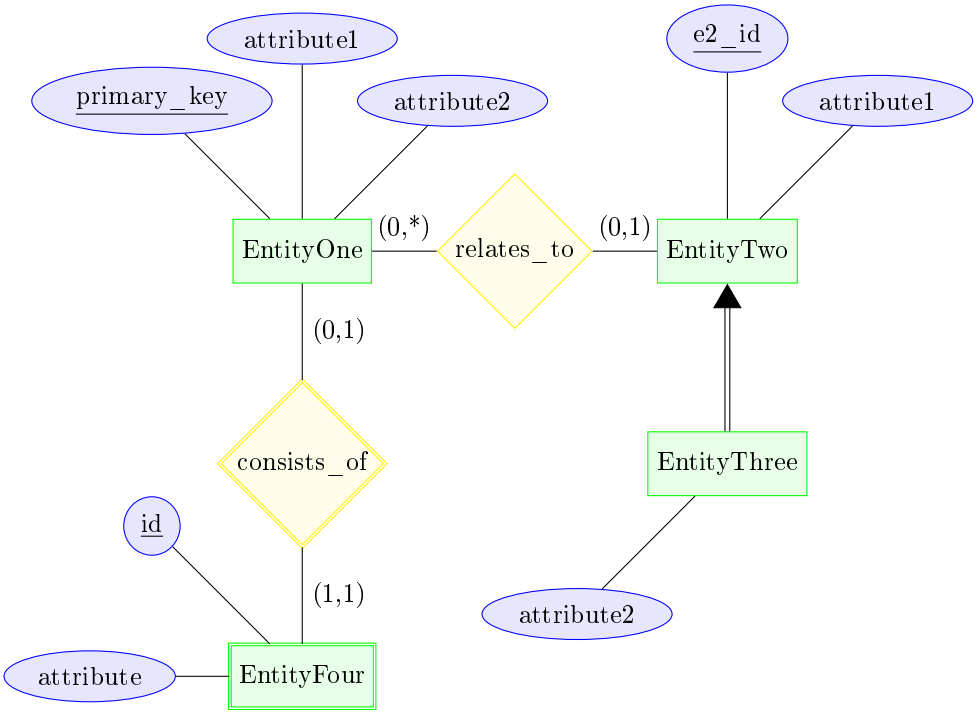
Bibliography

We use the `biblatex` package with biber[1] as backend, which most distributions already include. Your references are stored in the `report.bib` file and used with the `\cite` command. For example: [1]. If the previous example shows `kopka1995guide -` you need to re-run the compilation in the following order: pdflatex, biber, pdflatex, pdflatex. In case you need some help with references in latex, there is an excellent cheatsheet[2].

[1] `http://biblatex-biber.sourceforge.net/`
[2] `http://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf`

Tikz Drawing

Yon can also integrate ER diagrams using LaTeXand Ti*k*Z (instead of importing the PDF).



Allowlist

Below you will find a *Allowlist* of all available latex packages. These packages should be sufficient for the creation of your document. If other packages are necessary, this must be clarified with the tutor in advance.

```
%% Internationalization Latex Settings
\usepackage[ngerman,english]{babel}
\selectlanguage{english}

% Choose the encoding of the input text.
\usepackage[utf8]{inputenc}
% To choose the font encoding of the output text.
\usepackage[T1]{fontenc}

%% Packages for Bibliography
\usepackage{csquotes}
% Recommended for biblatex
\usepackage{xpatch}
% pdflatex -> biber -> pdflatex (2x)
\usepackage[backend=biber,style=numeric,language=english]{biblatex}

%% Lists
\usepackage{mdwlist}
\usepackage{paralist}
\usepackage{enumitem}
\renewcommand\labelitemi{--}

%% Floats
\usepackage{float}
% Adds Float |FloatBarrier beyond which floats may not pass
\usepackage[section]{placeins}

%% Images
% Allows you to insert graphic files within a document.
\usepackage{graphicx}
\usepackage[justification=centerlast, font=small]{caption}
\usepackage{subcaption}
%|usepackage[caption=false]{subfig}

%% Tables
\usepackage{array}
\usepackage{tabularx}
\usepackage{booktabs}
% For Tables longer than one Page
\usepackage{longtable}
\usepackage{multirow}
\usepackage{makecell, boldline}
```

```
%% Math
\usepackage{mathtools}
\usepackage{amsfonts}
%\usepackage{amsthm}% Incompatible with document
\usepackage{amssymb}
\usepackage{xfrac}

%% Code
% For Algorithms
\usepackage{algorithmic}
\usepackage{algorithm}

% Plain code printing
\usepackage{listings}
\usepackage{verbatim}

%% For Colors (\textcolor etc.)
\usepackage{color}
\usepackage{xcolor}

%% For Tikz
\usepackage{tikz}
% A library providing ER prefabs
\usetikzlibrary{er}
\usetikzlibrary{positioning,arrows.meta}

%% References
% For URLs
\usepackage{xurl}
% Refs
\usepackage{appendix}
\usepackage{varioref}
\usepackage[
colorlinks,          % Links without boarder
linkcolor=black,     % Color internal links
filecolor=black,     % Color external links
citecolor=black,     % Cite Color
urlcolor=black       % URL Color
]{hyperref}
\usepackage{cleveref}

%% Import
% For input external files
\usepackage{import}
% Import of external PDF's
\usepackage{pdfpages}

%% Further tools and styles
% For dummy text generation
\usepackage{lipsum}
```

```
% Support for setting the spacing between lines
\usepackage{setspace}
% For Todo Notes
\usepackage[colorinlistoftodos]{todonotes}
% For curly underline
\usepackage[normalem]{ulem}
```

References

[1] H Kopka and PW Daly. “A Guide to {\LaTeX}–Document”. In: (1995).